

INDEPENDENT RESEARCH PORTFOLIO / REPRODUCIBLE COLLIDER DATA ANALYSIS

Reconstructing the Z-Boson Peak from CERN Open Data: A Reproducible Dimuon Analysis

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*An independent reconstruction of the CMS dimuon spectrum, built around provenance,
data-quality auditing, and bounded inference.*

Abstract

A visible resonance is not yet a measurement: its interpretation depends on provenance, event selection, kinematic reconstruction, detector response, and the adequacy of the statistical model. This study reconstructs the dimuon invariant-mass spectrum from 83,761 preselected CMS Run2011A events released as CERN Open Data record 5202. The analysis verifies the file checksum and documented selection, recomputes each event's mass from the two muons' transverse momentum, pseudorapidity, and azimuth, and audits the redundant Cartesian four-vector fields. The primary reconstruction correlates with the supplied mass at 0.999999895, whereas 2,998 events show at least one material discrepancy between redundant coordinate representations under the stated consistency tolerance. A binned Voigt-plus-exponential fit to 12,108 events from 70 to 110 GeV returns a descriptive peak parameter of 90.7392 ± 0.0258 GeV; 200 bootstrap resamples give a 95% percentile interval of 90.6877-90.7850 GeV. The peak remains stable across tested windows and bin widths, but a Pearson chi-squared per degree of freedom of 3.893 rejects any precision interpretation of the compact line-shape model. The result reproduces the central model-to-observable-to-evidence chain of collider analysis while remaining deliberately bounded: the education-derived file lacks the luminosity, trigger, efficiency, calibration, simulation, and systematic information required for a cross section, electroweak parameter extraction, or competitive Z-mass measurement.

Keywords: CERN Open Data; CMS; Z boson; dimuon invariant mass; reproducible analysis; data quality; resonance fitting

1. Introduction

The Z-boson peak is among the most recognizable distributions in collider physics. In a histogram of opposite-sign dilepton invariant mass, a concentration of events near 91 GeV rises above a continuum and gives a direct visual connection between reconstructed tracks and an unstable intermediate particle. That apparent simplicity is deceptive. The peak shown on a page is the endpoint of choices about which collisions were retained, how particle momenta were represented, which calibrations were applied, how detector resolution and radiation were modeled, and what statistical quantity was fitted. A graph can make the Z visible without establishing that its mass, width, production rate, or significance has been measured.

This paper uses a deliberately compact dataset to examine that distinction. CERN Open Data record 5202 distributes 83,761 events selected from the 2011 CMS SingleMu primary dataset, with two muon candidates and a supplied dimuon mass in each row [1]. Its parent record describes the file as one of several Run2011A samples prepared for education and outreach [2]. The format removes much of the technical overhead of the full CMS analysis environment, but it also removes information. The portal explicitly states that the file contains only a subset of the event record and is unsuitable for a full physics analysis [1]. That warning is not a ceremonial disclaimer. It determines what questions the data can answer.

The physical setting remains authentic. The CMS detector combines a silicon tracker inside a 3.8 T solenoid with dedicated muon systems outside the calorimeters, allowing charged-particle trajectories and muon momenta to be reconstructed over a broad angular range [3]. Opposite-sign muon pairs receive contributions from the Drell-Yan process, including virtual-photon and Z-boson exchange, as well as lower-mass resonances. Full CMS measurements of the Drell-Yan spectrum use this same invariant-mass

logic, but add luminosity normalization, efficiency and acceptance corrections, detector simulation, radiation treatment, background estimation, and systematic uncertainties [4]. The education file therefore preserves the conceptual center of the analysis while withholding many of the layers required for a publishable measurement.

The research question is consequently narrower and more useful than whether the Z can be "found" in a public CSV: **What can a transparent analysis of a simplified CMS dimuon dataset genuinely reconstruct about the Z signal, and which parts of a publication-grade collider measurement remain outside the dataset's scope?** The answer developed here is that measured muon coordinates can be transformed into Lorentz-invariant mass, the source selection and data-quality audit can be made explicit, and a bounded signal-plus-background model can quantify a stable detector-level peak. Those achievements reproduce the central evidential chain of collider physics. They do not amount to a new Z-boson measurement because the sample lacks the information required to relate selected counts and fitted shape parameters to a calibrated physical observable.

Two features distinguish the present work from a standard histogram exercise. First, the supplied mass column is not accepted as an unexplained answer. Each event's mass is independently recomputed from transverse momentum, pseudorapidity, and azimuth after assigning the accepted muon rest mass. Second, the file's redundant coordinate systems are tested against one another. This comparison reveals that the cylindrical variables reproduce the supplied mass extremely closely, while a minority of stored energy and Cartesian momentum components exhibit material inconsistencies. The discrepancy does not invalidate the primary reconstruction, but it demonstrates why redundant fields should be treated as testable representations rather than interchangeable ground truth.

The paper then fits the Z region with a Voigt signal and exponential background, subjects the fitted location to bootstrap and analysis-choice perturbations, and evaluates goodness of fit. The combination is important. A stable parameter can still come from an inadequate model, and a small covariance uncertainty can still omit the dominant uncertainties. Here the peak is stable, but the pull structure and elevated chi-squared show that the model is descriptive rather than precision-grade. The strongest result is therefore not a decimal expansion of the Z mass. It is a fully inspectable chain from an official public record to a claim whose limits are stated as carefully as its numerical content.

2. Dataset, Provenance, and Analytical Contract

2.1 Source identity

The sole collision-data input is `Dimuon_SingleMu.csv` from CERN Open Data record 5202 [1]. The record identifies the source as data recorded in 2011 and published in 2019, derived from the CMS SingleMu primary dataset. The downloaded file contains 83,761 rows, occupies 11,806,904 bytes, and has Adler-32 checksum `3f2809bc`. The local analysis verifies that checksum before processing. Record 5202 is linked to parent record 545, which documents the broader family of Run2011A education samples [2]. Both records release the data under CC0 and state that CERN and CMS do not endorse independent works produced from them.

The parent record matters because it distinguishes two superficially similar samples. Its Zmumu selection requires two muons with transverse momentum above 20 GeV, $|\eta| < 2.1$, and mass between 60 and 120 GeV. The Dimuon selection used here is broader: two muons with $|\eta| < 2.4$,

at least one classified as a global muon, opposite electric charges, and supplied invariant mass between 0.3 and 300 GeV [2]. The broad sample can display low-mass and Z-region structure in one spectrum, but it is already filtered around a dimuon topology. The present analysis does not begin from all recorded collisions, and the appearance of a Z peak cannot be treated as an unbiased discovery exercise.

2.2 Row grain and schema

Each row represents one preselected dimuon event and contains 21 columns. `Run` and `Event` identify the collision record. For each muon, the file provides a type label (G for global or T for tracker), charge, transverse momentum `pt`, pseudorapidity `eta`, azimuth `phi`, energy `E`, and Cartesian momentum components `px`, `py`, and `pz`. The final column, `M`, is the supplied invariant mass of the pair [1]. These fields encode the same kinematics redundantly: `pt`, `eta`, and `phi` should determine the three-momentum, while `E`, `px`, `py`, and `pz` should form a compatible four-vector after the particle mass is specified.

Redundancy creates an opportunity for validation. If a paper simply plots `M`, a transcription or representation problem can pass unnoticed. If it reconstructs mass independently from both coordinate systems, agreement becomes an empirical property of the file. Finite decimal precision guarantees small residual differences, so exact bitwise equality is not expected. Material tails, however, require an explicit analytical response.

Table 1. Dataset identity and pre-analysis contract.

Item	Locked value	Analytical consequence
CERN record	5202; parent 545	Claims apply only to this education-derived file.
Local file	Dimuon_SingleMu.csv	One preselected dimuon event per row.
Size and checksum	11,806,904 bytes; Adler-32 3f2809bc	The analyzed bytes are traceable to the portal record.
Distributed rows	83,761	Counts refer to the CSV, not the complete SingleMu dataset.
Source selection	Opposite charge; two muons; $\text{abs}(\text{eta}) < 2.4$; at least one global; $0.3 < M < 300$ GeV	No independent discovery selection or significance claim.
Primary coordinates	<code>pt</code> , <code>eta</code> , <code>phi</code>	Used only after quantitative agreement with supplied <code>M</code> is established.
Prohibited outputs	Cross section, calibrated Z mass or width, weak mixing angle, discovery significance	Required information is absent from the education file.

2.3 Interpretation fixed before fitting

The analytical contract was set before the Z-window result was interpreted. Every row would be retained if the documented structural and selection checks passed. The primary mass would be computed from `pt`, `eta`, and `phi` only if that representation reproduced the supplied `M` across the sample. The stored Cartesian components would be audited separately rather than silently substituted. No event would be removed merely because the redundant representations disagreed; such a cut would create an unvalidated selection unless the discrepancy's origin were understood.

Most importantly, the fitted location parameter would not be labeled a measurement of the physical Z mass. CERN's education guide places simplified CSV files in histogramming, spreadsheet, and introductory Jupyter workflows [5]. The separate research guide identifies trigger configuration and

prescales, validated-run selection, luminosity, condition data, simulation, and experiment software as components of research-oriented analysis [6]. Their absence is therefore a known boundary, not a defect discovered after an inconvenient result. The paper can test whether an internally coherent and reproducible resonance reconstruction is possible. It cannot replace the experimental information that the source intentionally omits.

3. From Muon Coordinates to Invariant Mass

3.1 Event-level reconstruction

A muon's reconstructed four-momentum is written as $p^\mu = (E, p_x, p_y, p_z)$. The CSV's cylindrical variables are converted to Cartesian momentum through

$$p_x = p_T \cos\phi, p_y = p_T \sin\phi, p_z = p_T \sinh\eta,$$

where p_T is transverse momentum, ϕ is azimuth, and η is pseudorapidity. Assigning the muon rest mass $m_\mu = 0.1056583755$ GeV gives the positive-energy relation

$$E^2 = p_T^2 \cosh^2\eta + m_\mu^2, E > 0.$$

For two muons, four-momentum conservation defines the pair invariant mass:

$$m_{\mu\mu}^2 = (p_1 + p_2)^2 = (E_1 + E_2)^2 - |p_1 + p_2|^2.$$

The result is Lorentz invariant: analysts in longitudinally boosted frames obtain the same value. That property makes invariant mass particularly useful at a proton collider, where the momentum fractions carried by the colliding partons vary from event to event. Individual collisions do not display a particle's lifetime directly. Instead, many reconstructed pairs accumulate into a mass distribution whose localized structure can indicate a common intermediate state.

3.2 Why a peak appears

At the LHC, quark-antiquark annihilation can produce a virtual photon or Z boson that decays to a positive and negative muon. The resulting Drell-Yan mass spectrum contains a continuum and a strong enhancement in the Z region. CMS has used dimuon and dielectron spectra to measure the Drell-Yan differential cross section over a much wider mass range [4]. The education file also contains low-mass structures associated with heavy quarkonium families. These features make the broad spectrum informative, but only the Z region is modeled quantitatively here. The labels at the J/psi and Upsilon(1S) reference masses orient the reader; they are not substitutes for separate line-shape fits or state-by-state spectroscopy.

The observed Z profile is not the natural line shape alone. The physical propagator contributes a Breit-Wigner-like form; detector momentum resolution broadens and deforms the reconstructed mass; final-state photon radiation can move events toward lower masses; selection acceptance changes across kinematics; and residual backgrounds vary with mass. CMS muon-performance studies explicitly measure reconstruction and trigger efficiencies, momentum scale, and momentum resolution using collision data and calibrated resonance samples [7]. A simplified file that does not preserve the required calibration and simulation metadata cannot disentangle these contributions.

For this reason the present fit uses a Voigt profile, the convolution of a Lorentzian and a Gaussian, as a compact approximation. The Lorentzian half-width is fixed to half of the 2025 Particle Data Group total-width reference, while a Gaussian scale is fitted to absorb an effective detector term. The PDG listing gives $m_Z=91.1880\pm0.0020$ GeV and $\Gamma_Z=2.4955\pm0.0023$ GeV [8]. It also notes that the quoted mass corresponds to a specific mass-dependent Breit-Wigner convention. The fitted Voigt location in this paper is not the same estimator. Comparing them is useful as a scale and calibration diagnostic, but subtracting them does not create a precision test of the Standard Model.

3.3 Observable, estimator, and claim

Three objects must remain distinct. The **observable** is the reconstructed dimuon mass in each selected row. The **estimator** is the location parameter returned by a specified binned model over a specified range. The **physical parameter** is the convention-defined Z mass inferred by precision combinations after calibration and systematic treatment. Much overclaiming occurs when the second is renamed as the third. This paper reports all three levels explicitly: the event calculation, the descriptive fit, and the external reference. Their separation allows a visible Z signal to be treated as genuine evidence without granting the compact estimator authority it does not possess.

4. Reproducible Methods

4.1 Source lock and computational environment

Reproducible CMS open-data work begins by fixing the exact input and the analyst-owned environment in which it is interpreted [9]. The raw CSV was downloaded from the file URL recorded in `data/dataset_manifest.yaml`. Before analysis, the downloader verifies the expected byte count and Adler-32 checksum. Raw data are treated as immutable. The executable analysis is implemented once in `analysis/core.py` and called by both a command-line script and the Jupyter notebook, avoiding two implementations that could diverge. The recorded environment uses Python 3.9.6, NumPy 2.0.2, pandas 2.3.3, SciPy 1.13.1, and Matplotlib 3.9.4. NumPy supplies vectorized numerical operations and seeded resampling [10]; pandas handles table ingestion and validation [11]; SciPy supplies the Voigt profile and bounded nonlinear fit [12]; and Matplotlib produces the figures [13]. Exact versions are preserved in `requirements.txt` and the machine-readable result summary.

This arrangement follows a broader lesson from CMS open-data reuse: public availability of bytes does not by itself make an analysis usable. External analysts still own substantial parts of environment reconstruction, validation, workflow design, and documentation [9]. The notebook is therefore an inspectable interface to the analysis, not the sole computational record. Its outputs can be regenerated from the same core module, while JSON and CSV products permit independent checking without scraping values from plots.

4.2 Structural and selection audit

The first pass tests 11 conditions before physics interpretation: expected row count; exact 21-column schema; absence of missing values; absence of exact duplicate rows; uniqueness of the Run/Event pair; finiteness of all numeric cells; opposite-sign charges; at least one global muon; acceptance within the rounded boundary $|\eta| \leq 2.4$; supplied mass inside the portal range; and positive transverse momenta. The local acceptance test permits equality because the CSV exports four decimal places. Values selected with $|\eta| < 2.4$ upstream can therefore appear as 2.4000 after rounding.

No quality check is allowed to become an undocumented physics selection. Structural failures would stop the analysis. Representation discrepancies are instead measured and reported, because deleting affected events without a causal model could bias the sample.

4.3 Independent mass and representation audit

The primary invariant mass is reconstructed from pt , η , and ϕ using the equations in Section 3. It is compared event by event with the supplied M . A second mass is computed directly from stored E , p_x , p_y , and p_z , clipping negative mass-squared values to zero only for the purpose of defining a displayable mass while separately counting those negatives.

Coordinate consistency is also checked for each muon. Cartesian p_x and p_y are converted back to transverse momentum and azimuth; p_z and transverse momentum are converted to pseudorapidity; and a mass-shell energy is calculated. An event is flagged by the main diagnostic if either muon differs by more than 1% in transverse momentum or energy, or by more than 0.01 in pseudorapidity or azimuth. A severe diagnostic uses 10% and 0.10 thresholds. These tolerances are analytical alarms, not detector-quality definitions. Their purpose is to identify material disagreements far larger than ordinary decimal rounding.

4.4 Spectrum construction

The broad spectrum uses 299 geometrically spaced bins between 0.3 and 300 GeV. Because logarithmic bins have unequal widths, counts are divided by each bin width and plotted as events per GeV. Both axes are logarithmic, allowing low-mass structures and the Z region to share one panel without implying equal statistical precision. The analysis records orientation-window counts for 2.8-3.4 GeV, 8-12 GeV, 60-120 GeV, and 80-100 GeV. These windows summarize occupancy; they are not selection efficiencies or resonance yields.

4.5 Z-window model

The primary fit uses reconstructed cylindrical mass from 70 to 110 GeV in 0.5 GeV bins. For bin center m_i and width Δm , the expected count is

$$\lambda_i = N_S V(m_i; \mu, \sigma, \Gamma_Z/2) \Delta m + B \exp[k(m_i - m_Z)] \Delta m,$$

where V is a unit-area Voigt profile, N_S is the signal normalization, μ is its location, σ is the fitted Gaussian scale, and $\Gamma_Z/2 = 1.24775$ GeV is fixed. The background has density B at the PDG mass reference and exponential slope k . All five free parameters are bounded to physically usable ranges.

The model is fitted to binned counts by weighted nonlinear least squares with per-bin scale $\sqrt{\max(n_i, 1)}$. Reported parameter uncertainties are square roots of the covariance diagonal under that fitting model. Pulls use $(n_i - \lambda_i) / \sqrt{\max(\lambda_i, 1)}$, and Pearson chi-squared is the sum of the squared pulls. With 80 bins and five

fitted parameters, the primary goodness-of-fit statistic has 75 degrees of freedom. This is a transparent descriptive procedure, not an unbinned likelihood or a complete detector-response model.

4.6 Stability and resampling

Analysis-choice sensitivity is tested with six configurations: ranges 65-115, 70-110, 75-105, and 80-100 GeV; and bin widths 0.25, 0.5, and 1.0 GeV within the primary range. These variants probe whether the location is driven by one edge or binning choice. They do not span all plausible line shapes and therefore do not define a systematic uncertainty.

A nonparametric bootstrap resamples the 12,108 primary-range events with replacement, refits each sample, and records the location parameter. Two hundred repetitions are run with fixed random seed 20260710; all converge. The 2.5th and 97.5th percentiles form a resampling interval. Because every bootstrap sample inherits the same selection, coordinates, calibration state, and fit model, the interval estimates sampling variation under the existing pipeline only. It cannot cover omitted detector or model systematics.

5. Results

5.1 Structural integrity and selected sample

All 11 core checks pass. The file contains the expected 83,761 rows and 21 columns, with run numbers from 160957 to 173692 and supplied mass from 0.3001 to 296.812 GeV. There are no missing cells, nonfinite numeric values, exact duplicate rows, or repeated `Run/Event` keys. Every event has opposite-sign charge, positive transverse momentum, and at least one global muon. Four events contain a rounded `|eta|` value of exactly 2.4000; treating the exported boundary as `<=2.4` retains them consistently with finite-precision storage.

The source selection is also visible in the composition. There are 49,019 global-tracker pairs and 34,742 global-global pairs. Charge ordering is nearly balanced: 42,614 rows list the positive muon first and 41,147 list the negative muon first. These counts do not establish detector efficiency, but they help show that the file's schema and selection semantics are internally coherent.

The broad mass distribution contains 7,622 events in 2.8-3.4 GeV, 7,049 in 8-12 GeV, 12,768 in 60-120 GeV, and 11,142 in 80-100 GeV. Figure 1 shows why invariant-mass analysis is statistical rather than event-pictorial. Individual rows carry only two reconstructed muons; recognizable particle structure emerges when many rows occupy a common mass region.

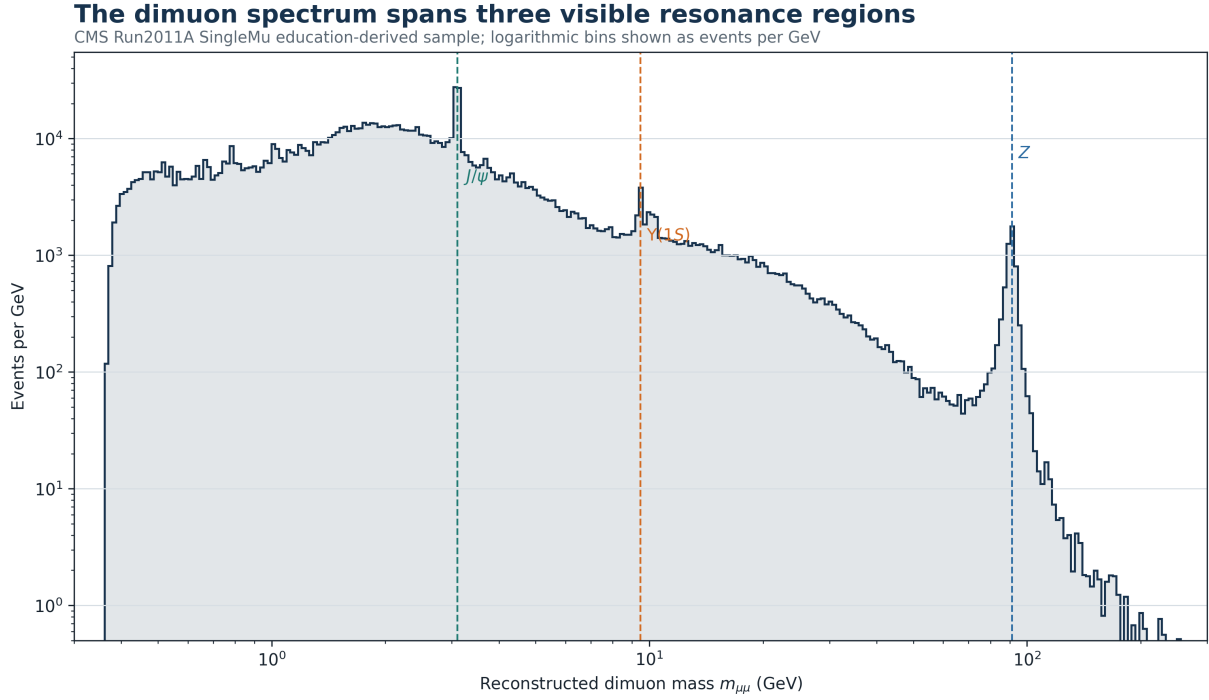


Figure 1. Full reconstructed dimuon spectrum. Mass is recomputed from pt, eta, and phi; geometrically spaced bins are normalized to events per GeV. Dashed lines mark reference masses for orientation. The labels do not represent separate fits.

5.2 Independent mass reconstruction

The cylindrical reconstruction strongly validates the primary event calculation. Across all 83,761 rows, its Pearson correlation with supplied M is 0.999999895. The median absolute difference is 0.00665 GeV, the 95th percentile is 0.03728 GeV, and 99.611% of events agree within 0.1 GeV. Agreement is tighter in the broad Z window: the median absolute difference is 0.000557 GeV and the 95th percentile is 0.002360 GeV. The residuals are small compared with the fitted peak scale and are compatible with the finite precision of the exported cylindrical variables.

The direct Cartesian calculation has a different tail. Its median absolute difference from M remains small at 0.00777 GeV, so most rows are consistent, but 403 events differ by more than 1 GeV and 50 differ by more than 10 GeV. Six rows produce negative pair mass squared before clipping. The component-level audit flags 2,998 events, or 3.579% of the file, above the combined 1%/0.01 tolerance and 711 events, or 0.849%, above the severe 10%/0.10 tolerance. The discrepancy is asymmetric between the two stored muon slots and is driven mainly by Cartesian-to-pt disagreement in the flagged tail. Within 60-120 GeV, only 91 events, 0.713% of the window, exceed the main tolerance.

Figure 3 makes the analytical decision visible. The blue cumulative curve for cylindrical reconstruction rises rapidly toward complete agreement; the orange Cartesian curve develops a long tail. The right panel shows the very narrow cylindrical residual distribution in the Z window. Consequently, the paper uses cylindrical coordinates for the physics result, retains all rows, and reports the Cartesian fields as a quality finding. It would be stronger neither to conceal the inconsistency nor to exclude events by an invented correction.

Mass reconstruction doubles as a data-quality test

The analysis uses the kinematic representation that reproduces the supplied mass across the sample.

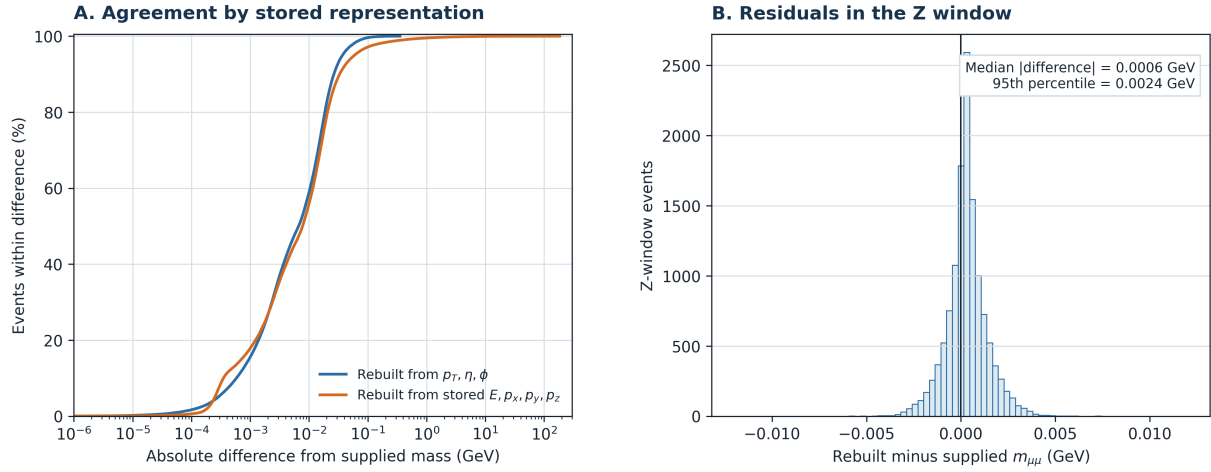


Figure 2. Validation of redundant mass representations. The left panel compares cumulative absolute discrepancies from supplied M ; the right panel resolves cylindrical-reconstruction residuals in the 60-120 GeV window. The long Cartesian tail motivates the primary-coordinate choice.

5.3 Primary Z-region fit

The 70-110 GeV fit contains 12,108 events. Its Voigt location parameter is 90.73924 GeV with covariance uncertainty 0.02576 GeV. The fitted Gaussian scale is 1.37769 $\pm$ 0.03368 GeV. Within the fit range, the model integrates to approximately 11,028 signal-component counts and 828 background-component counts, corresponding to a fitted signal fraction of 0.930. These are component integrals from the model, not truth labels assigned to individual collisions.

Table 2. Primary fit and stability diagnostics.

Quantity	Result	Interpretation
Primary range and bin width	70-110 GeV; 0.5 GeV	Fixed analysis configuration.
Events in fit range	12,108	Preselected events, not efficiency-corrected yield.
Voigt location	90.7392 $\pm$ 0.0258 GeV	Descriptive fit parameter; covariance uncertainty only.
Fitted Gaussian scale	1.3777 $\pm$ 0.0337 GeV	Effective broadening term, not a direct detector-resolution measurement.
Offset from PDG 2025 mass	-0.4488 GeV	Diagnostic comparison; estimators and calibration differ.
Bootstrap 95% interval	90.6877-90.7850 GeV	Sampling variation under the unchanged pipeline.
Tested configuration span	90.7315-90.7859 GeV	Limited range/binning sensitivity, not full systematics.
Pearson chi-squared/ndf	291.94/75 = 3.893	Compact model is inadequate for precision inference.

The location lies 0.44876 GeV below the 2025 PDG mass reference. That difference is much larger than the fit covariance uncertainty, but the comparison is not a valid significance calculation. The covariance describes local statistical curvature within the chosen model; it omits momentum-scale calibration, resolution variation, final-state radiation, acceptance, background-form uncertainty, and the

difference between the Voigt location and the PDG line-shape convention. The elevated goodness-of-fit statistic independently demonstrates that the model does not absorb all structure present in the data.

Figure 2 displays both strengths and weaknesses of the fit. The total curve captures the peak's broad location and scale, and the fitted background remains a small component in the chosen window. The pull panel, however, contains excursions and organized deviations rather than featureless unit-scale noise. Pearson chi-squared is 291.94 for 75 degrees of freedom, or 3.893 per degree of freedom. A model that were adequate under the stated uncertainties would ordinarily produce a value nearer one. The observed value does not identify one missing physical effect, but it decisively prevents the fit from being presented as a precision line-shape description.

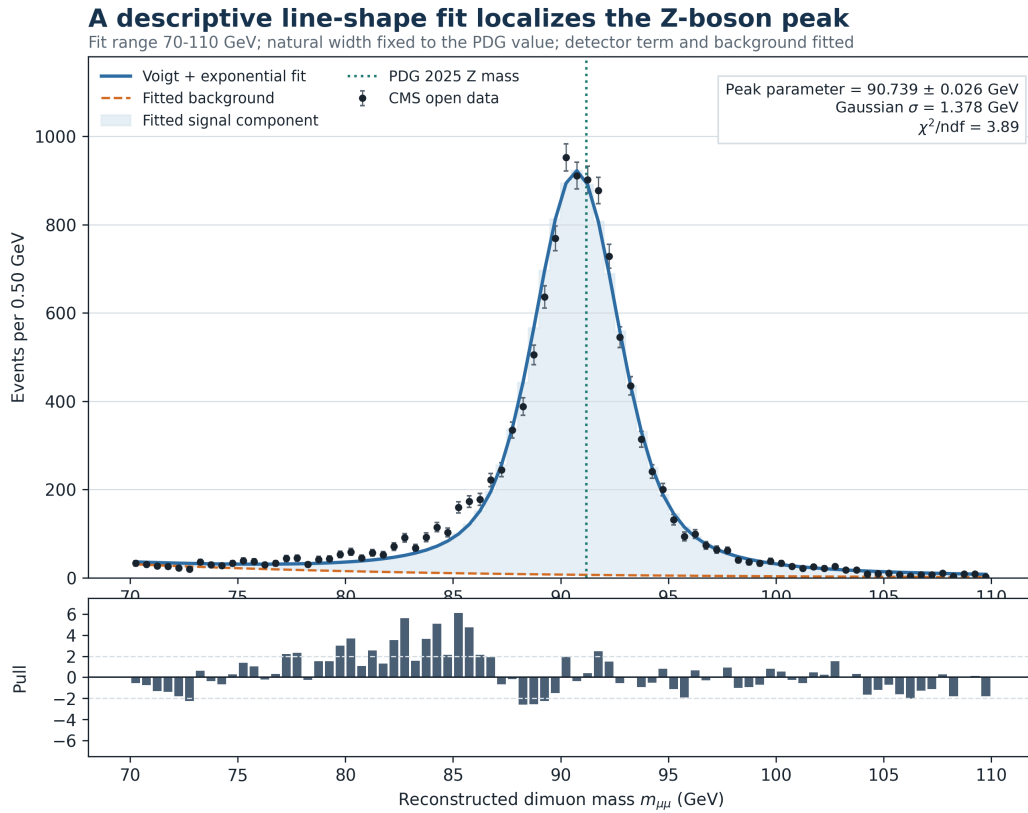


Figure 3. Descriptive Z-window fit and pulls. A Voigt signal with fixed PDG natural width and fitted Gaussian scale is combined with an exponential background. The location is stable, but structured pulls and chi-squared/ndf of 3.89 show that the model is not precision-grade.

5.4 Bootstrap and analysis-choice sensitivity

All 200 bootstrap fits converge. Their mean location is 90.73293 GeV, standard deviation 0.02690 GeV, and median 90.73368 GeV. The 95% percentile interval is 90.68771-90.78503 GeV. Its scale agrees with the covariance uncertainty, which indicates that the primary fit is numerically well behaved under event resampling. It does not repair the model inadequacy or include shared systematic shifts.

The six range-and-binning variants return locations from 90.73155 to 90.78594 GeV. Widening the range to 65-115 GeV gives the lower endpoint; narrowing to 80-100 GeV gives the upper endpoint. Within 70-110 GeV, changing the bin width from 0.25 to 1.0 GeV moves the location by only about

0.0054 GeV, although chi-squared per degree of freedom changes substantially because bin aggregation changes the residual accounting. Every tested configuration retains a chi-squared per degree of freedom well above one, from 2.49 to 7.03. Stability is therefore real but limited in meaning: the estimator repeatedly finds the same approximate peak under nearby choices, while every version remains too simple for precision inference.

6. What the Reconstruction Establishes

6.1 A successful detector-level reconstruction

Several conclusions are robust within the stated sample. The downloaded bytes match the official CERN record. The schema and documented preselection are reproduced locally. Independent four-vector construction from `pt`, `eta`, and `phi` yields an invariant mass that agrees with the supplied value across almost the entire file. The broad spectrum contains recognizable resonance structure, and a compact model repeatedly localizes the dominant high-mass peak near the expected Z scale. These are not simulated demonstration numbers: they arise from recorded CMS collision data, transformed by published code into a verifiable distribution.

The result also has explanatory value. It exposes the chain that a polished collaboration plot can conceal from a new analyst: track-derived coordinates become four-vectors; four-vectors become an invariant mass; event selections define which mass distribution exists; a histogram turns event-level quantities into population structure; a statistical model compresses that structure into parameters; diagnostics determine how far those parameters may be interpreted. The Z peak is persuasive because the links agree, not simply because one bin is tall.

6.2 Why the fitted location is not a Z-mass measurement

The fitted value, 90.7392 +/- 0.0258 GeV, should be read as a location parameter for this sample, range, binning, and model. Calling it a measurement of m_Z would imply that its uncertainty captures the important ways the estimator can differ from the physical parameter. It does not. The 0.4488 GeV offset from the PDG reference cannot be converted into a meaningful number of standard deviations by dividing by 0.0258 GeV, because that denominator is only the fit covariance uncertainty. The chi-squared result already shows that the statistical model is incomplete, and the source does not provide a momentum-scale uncertainty with which to calibrate the difference.

Multiple omitted effects could contribute in the same direction or in competing directions. CMS muon-performance work treats momentum scale and resolution as measured detector properties, varies them with kinematics, and uses calibrated resonance samples and simulation [7]. A full Drell-Yan measurement accounts for acceptance, efficiency, backgrounds, and final-state radiation [4]. The PDG mass itself is tied to a specified line-shape convention [8]. The compact Voigt model has no claim to isolate those effects. The defensible statement is therefore bounded: the offset is consistent with the limitations of an uncalibrated education sample and descriptive model, but this analysis cannot assign it to one cause.

The same logic applies to the fitted Gaussian scale of 1.3777 GeV. It summarizes the broadening needed by a one-parameter Gaussian convolution in this window. It is not a measurement of the CMS dimuon resolution, which varies with each muon's transverse momentum, pseudorapidity, reconstruction

category, and calibration state. Nor is the fixed Lorentzian input an extraction of the natural width. Fixing Γ_Z stabilizes the descriptive fit; it does not test that value.

6.3 Missing layers of a publication-grade result

A cross section requires a selected yield to be related to integrated luminosity and corrected for trigger, reconstruction, identification, and acceptance effects. Record 5202 supplies neither an analysis luminosity nor the trigger and prescale history needed to define one. It lacks per-event quality and condition information, full object content, simulated signal and background samples, and systematic variations. The CERN research guide treats these as distinct tasks in a research workflow [6]. Without them, an apparently precise selected-event count has no controlled conversion to an underlying production rate.

A discovery claim would require a predeclared signal region, a defensible background-only model, nuisance parameters, and a significance calculation that accounts for selection and model uncertainty. Here the source was intentionally selected to contain opposite-sign dimuons across a bounded mass range. The analysis knows in advance that a Z-rich region is present. Its exponential term is a local descriptive background, not a discovery-grade prediction.

An electroweak parameter extraction requires still more information. CMS's 7 TeV weak-mixing-angle analysis used the joint behavior of dilepton mass, rapidity, and decay angle, together with detector and systematic modeling [14]. A one-dimensional histogram from a reduced education CSV cannot reproduce that inference. The limitation is substantive rather than hierarchical: the present project answers a different question well.

6.4 What the representation discrepancy means

The Cartesian-field inconsistency is an important result precisely because it is easy to overstate. It shows that redundant values in this exported education artifact are not perfectly interchangeable. It does not establish a fault in CMS tracking, the muon system, or the original event reconstruction. The simplified CSV has passed through a derivation and rounding process not reconstructed here, and most rows remain mutually consistent. The asymmetry between muon slots suggests a transformation or export-level pattern worth further forensic study, but the current evidence does not identify its source.

Using the well-validated cylindrical representation is a proportionate response. The primary mass agrees extremely closely with supplied M , the relevant Z-window mismatch rate is small, and no unsupported deletion is introduced. Reporting the discrepancy also improves future reuse: another analyst can choose to reproduce the current decision, inspect the flagged rows, or derive a correction from a better-documented source. Quietly mixing representations would make none of those options visible.

6.5 The bounded scientific claim

The analysis reconstructs a real Z-region signal in a preselected detector-level sample and quantifies where a simple model places it. It validates the primary kinematics, demonstrates numerical stability under limited perturbations, and rejects a precision reading through explicit goodness-of-fit evidence. It does not measure the Z production cross section, natural width, weak mixing angle, discovery significance, or world-average mass. This is not a retreat from the result. It is the result stated at the level the evidence can carry.

7. Reproducibility as a Scientific Result

Reproducibility is often reduced to whether code runs. For an analysis paper, three forms are needed. **Provenance reproducibility** asks whether another analyst can identify the exact input. **Computational reproducibility** asks whether the transformations and numerical outputs can be regenerated. **Inferential reproducibility** asks whether the boundary between results and interpretation is visible. A notebook can satisfy the second imperfectly while failing the first and third.

This project preserves the chain in Figure 4. The source manifest records the portal and parent records, file URL, byte count, checksum, license, and documented selection. The environment file pins package versions. The core module implements calculations, audits, fitting, resampling, and figure generation; the command-line entry point and executed notebook call that same module. Machine-readable outputs include the complete analysis summary, quality checks, fit-bin values and model components, and the range/binning sensitivity table. The research files link claims to official sources or local artifacts. The manuscript then carries the interpretation boundary into the abstract, results, discussion, and conclusion.

From public record to bounded physics claim

Every arrow is a reproducibility checkpoint, not merely a plotting step.



The education dataset makes the inference chain visible while limiting the final claim.

Figure 4. Reproducibility and interpretation pipeline. The source, audit, reconstruction, spectrum, model, and final claim are separate checkpoints. A reproducible computation is incomplete if the last arrow permits a stronger claim than the preceding evidence.

The chain also exposes productive points of disagreement. An analyst can replace the binned least-squares fit with an unbinned likelihood, add a radiative-tail model, vary the background family, stratify resolution by muon kinematics, or inspect the anomalous Cartesian rows. Because parameters and intermediate tables are preserved, such changes can be compared against the existing baseline instead of rebuilding an undocumented plot. This is the practical value of open analysis: it makes assumptions editable.

There remains a hard ceiling. Exact software recreation cannot restore luminosity, trigger history, calibration conditions, or simulations that the source file does not contain. For a publication-grade extension, the appropriate next step is not to decorate the current fit with more decimal places. It is to move to a research-quality CMS Open Data release with validated runs, trigger and luminosity accounting, object selections, simulation, efficiency corrections, calibration variations, and a likelihood whose nuisance parameters represent the dominant uncertainties. The present artifact provides a tested analysis skeleton for that larger task while making clear which evidential layers must be added.

8. Conclusion

The Z-boson peak can be reconstructed cleanly from CERN's education-derived CMS dimuon data, but the most informative part of the exercise is not the peak alone. The source was locked by checksum, its documented selection was reproduced, and every event's mass was independently calculated from muon kinematics. That calculation agrees with the supplied mass at high precision. A separate audit uncovered a minority of material inconsistencies in the redundant Cartesian fields, demonstrating why even familiar public data should be tested before use.

The primary Voigt-plus-exponential fit places the detector-level peak at 90.7392 ± 0.0258 GeV and remains stable under the tested resampling, range, and binning changes. Yet its chi-squared per degree of freedom is 3.893, and its uncertainty omits the calibration, response, radiation, acceptance, background, and line-shape effects needed for a physical Z-mass inference. The fitted location is therefore a reproducible descriptive estimator, not a competitive measurement.

That distinction is the paper's central conclusion. The analysis successfully reproduces the collider-physics chain from recorded muon coordinates to population-level resonance evidence. It also shows where that chain stops when an education file lacks the metadata and models required for precision. A credible open-data paper is not one that makes the largest possible claim from public numbers. It is one in which another reader can trace the exact path from public source to bounded conclusion, rerun each step, and see why the result means no more and no less than the evidence permits.

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